Q1 JEE Main 2020 - 2 September (Morning)

On heating compound (A) gives a gas (B) which is a constituent of air. This gas when treated with H_2 in the presence of a catalyst gives another gas (C) which is basic in nature. (A) should not be:

(A) $Pb(NO_3)_2$
(B) $(NH_4)_2Cr_2O_7$
(C) NH_4NO_2
(D) NaN_3

Q2 JEE Main 2020 - 3 September (Morning)

Aqua regia is used for dissolving noble metals (Au, Pt, etc.). The gas evolved in this process is

(A) NO
(B) N_2
(C) N_2O_5
(D) N_2O_3

Q3 JEE Main 2020 - 3 September (Morning)

If the boiling point of H_2O is $373K$, the boiling point of H_2S will be

(A) less than $300K$
(B) more than $373K$
(C) equal to $373K$
(D) greater than $300K$ but less than $373K$

Q4 JEE Main 2020 - 3 September (Morning)

In a molecule of pyrophosphoric acid, the number of $P-OH, P=O$ and $P-O-P$ bonds / moiety(ies) respectively are

(A) 4,2 and 0
(B) 4,2 and 1
(C) 3,3 and 3
(D) 2,4 and 1

Q5 JEE Main 2020 - 4 September (Morning)

On heating, lead (II) nitrate gives a brown gas (A). The gas (A) on cooling changes to a colourless solid/liquid (B). (B) on heating with NO changes to a blue solid (C). The oxidation number of nitrogen in solid (C) is :

(A) $+3$
(B) $+4$
(C) $+5$
(D) $+2$

Q6 JEE Main 2020 - 4 September (Evening)

The reaction in which the hybridisation of the underlined atom is affected is

(A) $\underline{XeF}_4 + SbF_5 \rightarrow$

(B) $H_2\underline{SO}_4 + NaCl \xrightarrow{420K}$

(C) $H_3PO_2 \xrightarrow{\text{Disproportionation}}$

(D) $\underline{NH}_3 \xrightarrow{H^+}$

Q7 JEE Main 2020 - 5 September (Morning)

The structure of PCl_5 in the solid state is

(A) tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$
(B) square pyramidal
(C) trigonal bipyramidal
(D) square planar $[PCl_4]^+$ and octahedral $[PCl_6]^-$

Q8 JEE Main 2020 - 5 September (Evening)

Reaction of ammonia with excess Cl_2 gives

(A) NH_4Cl and HCl
(B) NCl_3 and HCl
(C) NCl_3 and NH_4Cl
(D) NH_4Cl and N_2

Q9 JEE Main 2020 - 6 September (Morning)

The correct statement with respect to dinitrogen is

(A) N_2 is paramagnetic in nature
(B) it can be used as an inert diluent for reactive chemicals
(C) it can combine with dioxygen at $25^{\circ}C$
(D) liquid dinitrogen is not used in cryosurgery

Q10 JEE Main 2020 - 6 September (Morning)

The number of $Cl = O$ bonds in perchloric acid is,

Q11 JEE Main 2020 - 6 September (Evening)

The reaction of NO with N_2O_4 at $250K$ gives

(A) N_2O_3
(B) N_2O_5
(C) N_2O
(D) NO_2

Q12 JEE Main 2020 - 6 September (Evening)

Reaction of an inorganic sulphite X with dilute H_2SO_4 generates compound Y. Reaction of Y with NaOH gives X. Further, the reaction of X with Y and water affords compound Z. Y and Z, respectively, are

(A) S and Na_2SO_3
(B) SO_2 and $NaHSO_3$
(C) SO_2 and Na_2SO_3
(D) SO_3 and $NaHSO_3$

Q13 JEE Main 2020 - 7 January (Morning)

Chlorine reacts with hot and concentrated $NaOH$ and produces compounds (X) and (Y). Compound (X)
gives white precipitate with silver nitrate solution. The average bond order between Cl and O atoms in (Y) is

Q14 JEE Main 2020 - 8 January (Morning)

The number of S-O bond in $S_2O_8{}^{2-}$ and number of S-S bond in Rhombic sulphur are respectively:

(A) $8,\ 8$
(B) $6,\ 8$
(C) $2,\ 4$
(D) $4,\ 2$

Q15 JEE Main 2020 - 8 January (Evening)

White phosphorus on reaction with concentrated $NaOH$ solution in an inert atmosphere of CO_2 gives phosphine and compound (X). (X) on acidification with HCl gives compound (Y). The basicity of compound $(Y)$is :

(A) 1
(B) 2
(C) 3
(D) 4

Q16 JEE Main 2020 - 8 January (Evening)

Arrange the following bonds according to their average bond energies in descending order :

(A) $C-Cl>C-Br>C-I>C-F$
(B) $C-F<C-Cl<C-Br<C-I$
(C) $C-F>C-Cl>C-Br>C-I$
(D) $C-I<C-Br<C-F<C-Cl$

Q17 JEE Main 2020 - 9 January (Morning)

The acidic, basic and amphoteric oxides, respectively, are :

(A) $MgO,\ P_4O_{10},\ Al_2O_3$
(B) $N_2O_3, Li_2O,\ Al_2O_3$
(C) $SO_3,\ Al_2O_3,\ Na_2O$
(D) $P_4O_{10},\ Al_2O_3,\ MgO$

Q18 JEE Main 2020 - 9 January (Morning)

X melts at low temperature and is a bad conductor of electricity in both liquid and solid state. X is :

(A) Hg
(B) ZnS
(C) SiC
(D) CCl_4

Answer Key

Q1 (A)	**Q2** (A)	**Q3** (A)	**Q4** (B)
Q5 (A)	**Q6** (A)	**Q7** (A)	**Q8** (B)
Q9 (B)	**Q10** (3)	**Q11** (A)	**Q12** (B)
Q13 (1.67)	**Q14** (A)	**Q15** (A)	**Q16** (C)
Q17 (B)	**Q18** (D)		

Q1

$2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$

$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2O_3$

$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

$2NaN_3 \xrightarrow{\Delta} 2Na + 3N_2$

Q2

$Au + 4H^+ + NO_3^- + 4Cl^- \rightarrow AuCl_4^- + NO + 2H_2O$

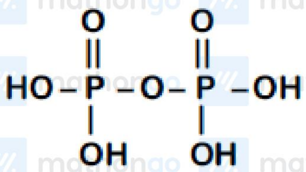
$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \rightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$

Q3

Boiling point of H_2S is $213K$.

Q4

Pyrophosphoric acid $(H_4P_2O_7)$



4 $P - OH$ bonds
2 $P = O$ bonds
1 $P - O - P$

Q5

$Pb(NO_3)_2 \xrightarrow{\Delta} PbO + \underset{\text{(brown gas)}}{NO_2} \uparrow$

A is NO_2

$2NO_2 \xrightarrow{\text{cooling}} N_2O_4$

$N_2O_4 + NO \longrightarrow N_2O_3$
oxidation state of N in C is +3

Q6

$xeF_4 + SbF_5 \rightarrow [XeF_3]^+[SbF_6]^-$
$sp^3d^2 \quad sp^3d$

$H_3PO_2 \xrightarrow{\text{Disproportionation}} sp^3H_3PO_4 + PH_3$

$NH_3 \xrightarrow{H^+} sp^3 \quad sp^3$
$sp^3 \quad NH_4^+$
sp^3

Q7

PCl_5 in solid state exist as $[PCl_4]^+[PCl_6]^-$
$[PCl_4]^+$ is tetrahedral
$[PCl_6]^-$ is octahedral

Q8

$$NH_3 + \underset{\text{excess)}}{3Cl_2} \longrightarrow \underset{\text{(explosive)}}{NCl_3} + 3HCl$$

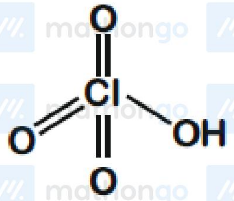
Q9

N_2 is diamagnetic in nature.

$N_2 + O_2 \xleftrightarrow{2000K} NO(g)$
Because of its inertness it is used where an inert atmosphere is required.

Q10

The structure of perchloric acid is



The number $\mathrm{Cl}=\mathrm{O}$ bonds in $\mathrm{HClO_4}$ is 3 .

Q11

$$2\mathrm{NO}+\mathrm{N_2O_4} \xrightarrow{250\mathrm{K}} \mathrm{N_2O_3}$$

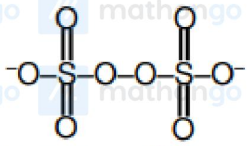
Q12

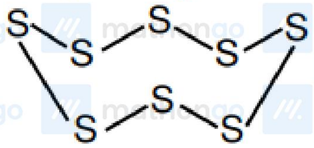
$$\mathrm{X}=\mathrm{Na_2SO_3}$$
$$\mathrm{SO_3^{2-}}+\mathrm{H_2SO_4} \rightarrow [\mathrm{H_2SO_3}] \rightarrow \mathrm{H_2O}+\mathrm{SO_2}$$
$$\mathrm{SO_2}+2\mathrm{NaOH} \rightarrow \mathrm{Na_2SO_3}+\mathrm{H_2O}$$
$$\mathrm{Na_2SO_3}+\mathrm{SO_2}+\mathrm{H_2O} \rightarrow 2\mathrm{NaHSO_3}$$

Q13

$$3Cl_2+\underset{\text{Hot \& conc}}{6NaOH} \rightarrow \underset{(X)}{5NaCl}+\underset{(Y)}{NaClO_3}+3H_2O$$
$$NaCl+AgNO_3 \rightarrow \underset{ppt.}{AgCl}+NaNO_3$$
$$\text{Y is } NaClO_3 \quad ClO_3^- \text{ (bond order)}=\frac{5}{3}=1.67$$

Q14

$S_2O_8^{2-}$ 

S_8 

Q15

$$P_4 + NaOH + H_2O \longrightarrow PH_3 + NaH_2PO_2 \xrightarrow{HCl} H_3PO_2$$

Q16

Bond energy $\propto \frac{1}{\text{Bond length}}$

Q17

Non-metal oxides are acidic in nature
alkali metal oxides are basic in nature
Al_2O_3 is amphoteric.

Q18

$CCl_4 \rightarrow$ Non-conductor in solid and liquid phase.